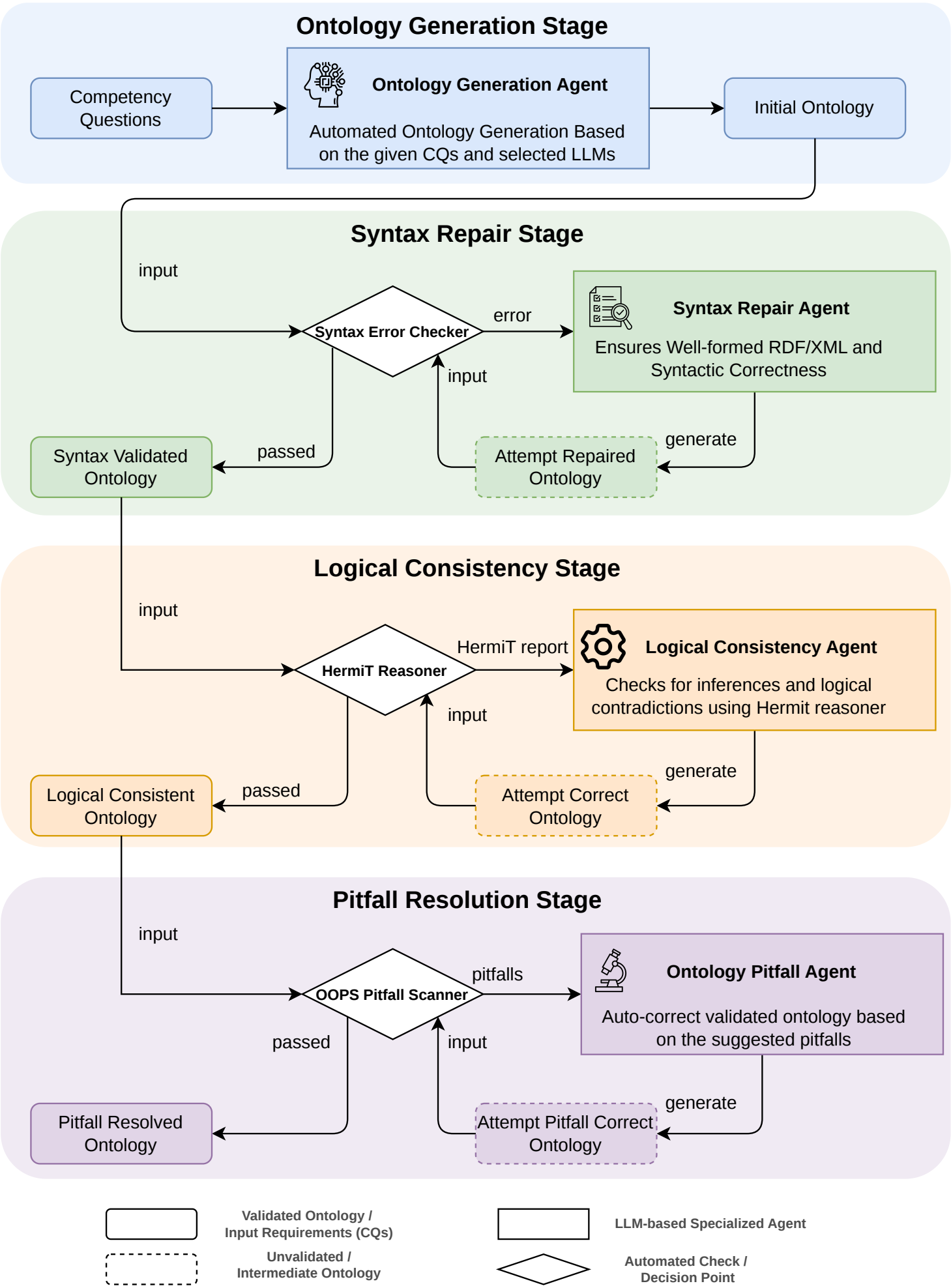
Ontology Generation Stage
Competency Questions
Ontology Generation Agent
Automated Ontology Generation Based on the given CQs and selected LLMs
Initial Ontology
Syntax Repair Stage
input
Syntax Error Checker
error
Syntax Repair Agent
Ensures Well-formed RDF/XML and Syntactic Correctness
input
generate
Syntax Validated Ontology
passed
Attempt Repaired Ontology
Logical Consistency Stage
input
HermiT Reasoner
HermiT report
Logical Consistency Agent
Checks for inferences and logical contradictions using Hermit reasoner
input
generate
Logical Consistent Ontology
passed
Attempt Correct Ontology
Pitfall Resolution Stage
input
OOPS Pitfall Scanner
pitfalls
Ontology Pitfall Agent
Auto-correct validated ontology based on the suggested pitfalls
passed
input
generate
Pitfall Resolved Ontology
Attempt Pitfall Correct Ontology
Validated Ontology / Input Requirements (CQs)
Unvalidated / Intermediate Ontology
LLM-based Specialized Agent
Automated Check / Decision Point